Supplementary Material for "News Publisher Reliability from Article Collections: A Publisher-Disjoint Evaluation"

Companion to the ECIR 2027 IR for Good submission: full error and bias analyses, and experimental details.

1 Model performance

Figure 1 gives the article-level confusion matrices per fold and on average.

2 Publisher-level Results

Figure 2 gives the publisher-level confusion matrices after majority voting.

Most errors are between adjacent levels. The most frequent is l_5 predicted as l_4 (first for BERT, Llama 3 and Mistral, second for RoBERTa): l_5 is the single score 100, l_4 the range 75–99. Errors of two levels still affect 39 to 46 publishers per model, mostly between l_2 and l_4; errors of three or more levels affect 10 to 17.

Majority-vote ties are broken toward the *lowest* level. This decides 13 of the 439 publishers for BERT, 20 for RoBERTa, 15 for Llama 3 and 11 for Mistral, always toward lower predicted reliability.

3 Political leaning

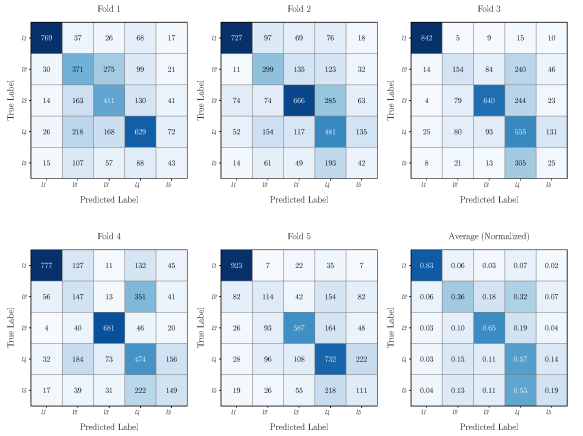
Figure 3 shows the article-level confusion matrices for each leaning.

- **Slightly Right** (largest group, $N = 7490$) and **Non labelled** are strongly diagonal; on Slightly Right accuracy is 66.4–69.0%.
- **Far Left** ($N = 1015$) reaches 28.1–29.5% accuracy, 36.5% for Mistral, and **Far Right** ($N = 2142$) 31.7–40.6%. Errors fall mostly above the diagonal (overestimated reliability) and come from few domains.
- **Slightly Left** errors fall mostly below the diagonal (underestimated reliability) for all four models.

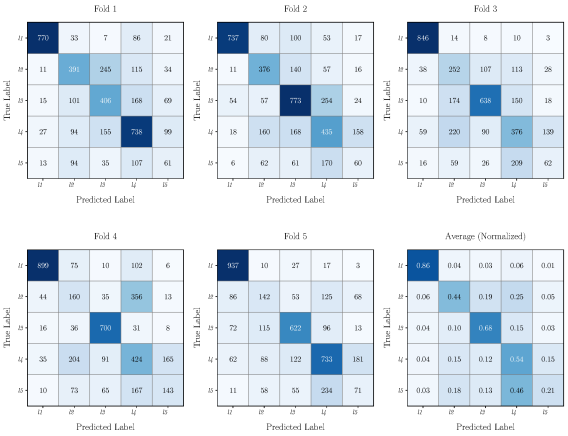
These matrices are descriptive: leaning and reliability are both publisher attributes and are associated in the corpus. Section 6 controls for this.

4 Error Direction at the Article Level

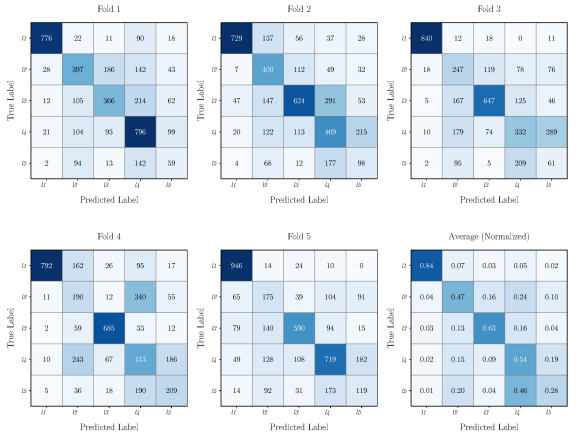
Figure 4 gives the Mean Error Direction (MED) per true reliability level, computed over misclassified articles only. Errors regress toward the centre of the scale:



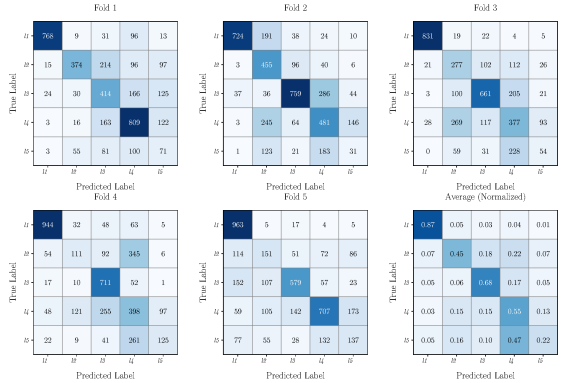
(a) BERT



(b) RoBERTa

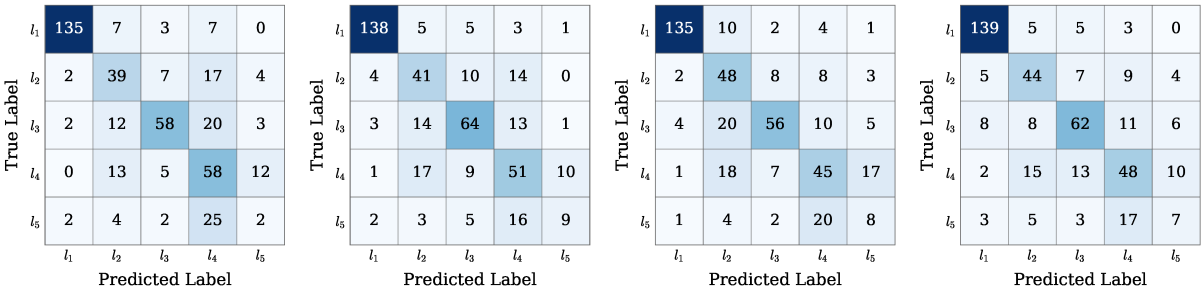


(c) Llama 3



(d) Mistral

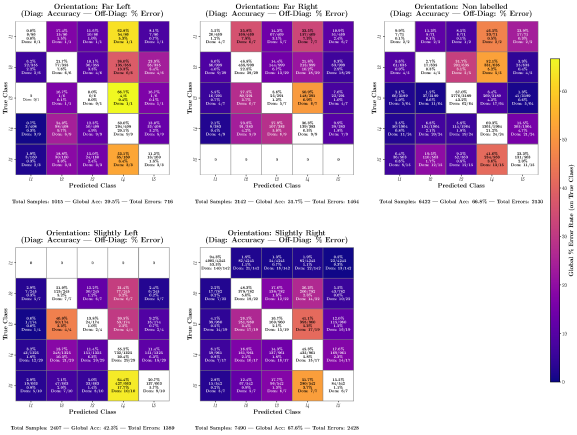
Figure 1: Article-level confusion matrices for each of the five folds and their average. Rows are true labels, columns predicted labels.



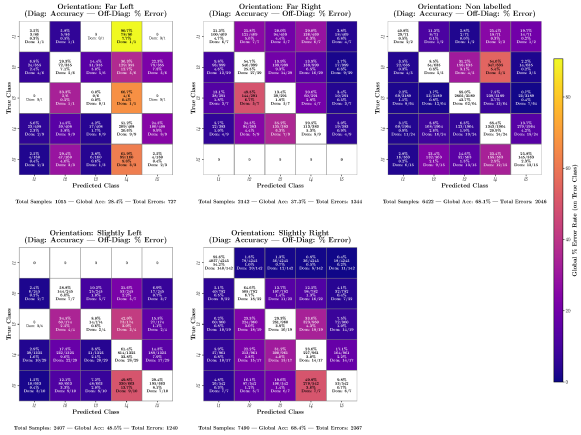
(a) BERT (b) RoBERTa (c) Llama 3 (d) Mistral

Figure 2: Publisher-level confusion matrices. Rows are true levels, columns predicted levels.

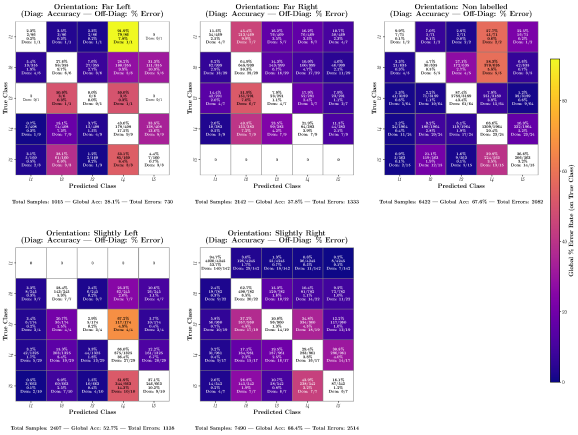
- l_1 **and** l_5 can only be over- and underestimated respectively. The MED is about +2.1 for l_1 and about −1.7 for l_5 (BERT −1.60, RoBERTa −1.72, Llama −1.67, Mistral −1.74); most l_5 errors land on l_4.
- l_3 has the smallest MED, 0.10 to 0.35: it absorbs overestimated l_2 and underestimated l_4 articles, and the two directions largely cancel.



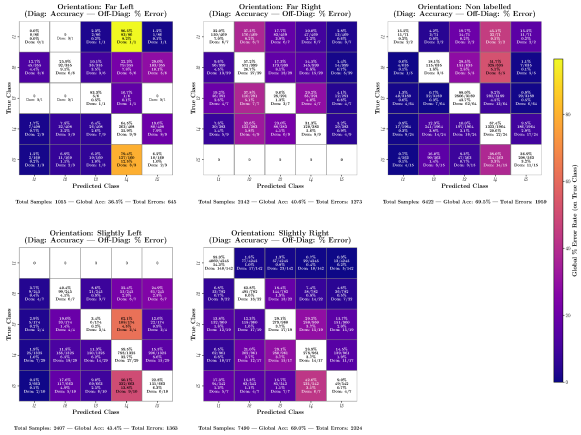
(a) BERT



(b) RoBERTa



(c) Llama 3



(d) Mistral

Figure 3: Confusion matrices stratified by political leaning. Rows are true labels, columns predicted labels. Cells show class-wise percentages and counts, dataset-level percentages and contributing domains ('Dom'); colour encodes the error rate per true class.

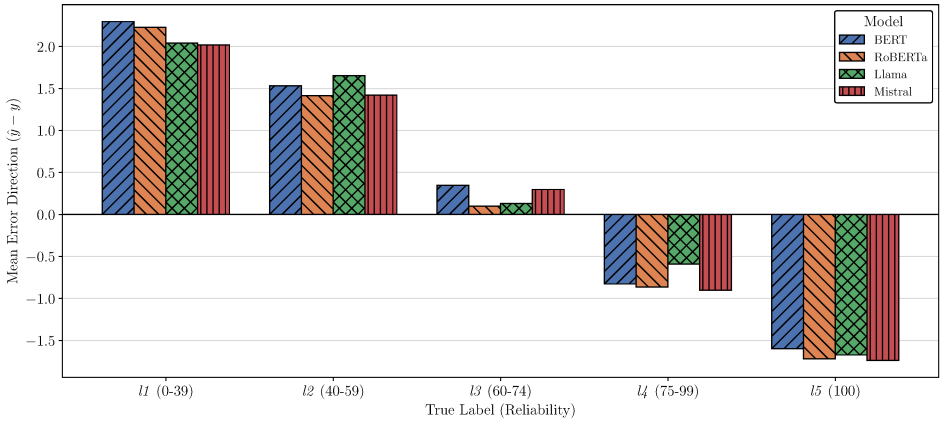
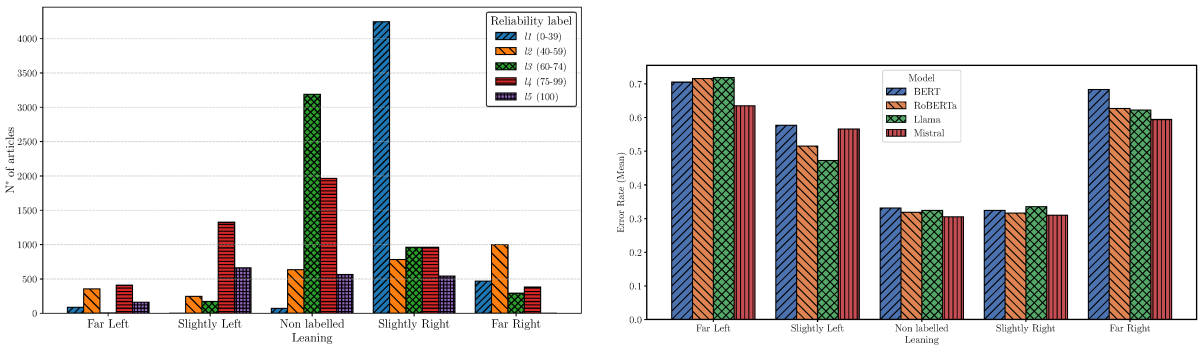


Figure 4: Mean Error Direction by true reliability level. Positive values indicate overestimation of reliability, negative values underestimation.



(a) Reliability labels across political leanings.

(b) Error rate by leaning and model.

Figure 5: Composition of the corpus by political leaning (left) and the resulting per-leaning error rates (right).

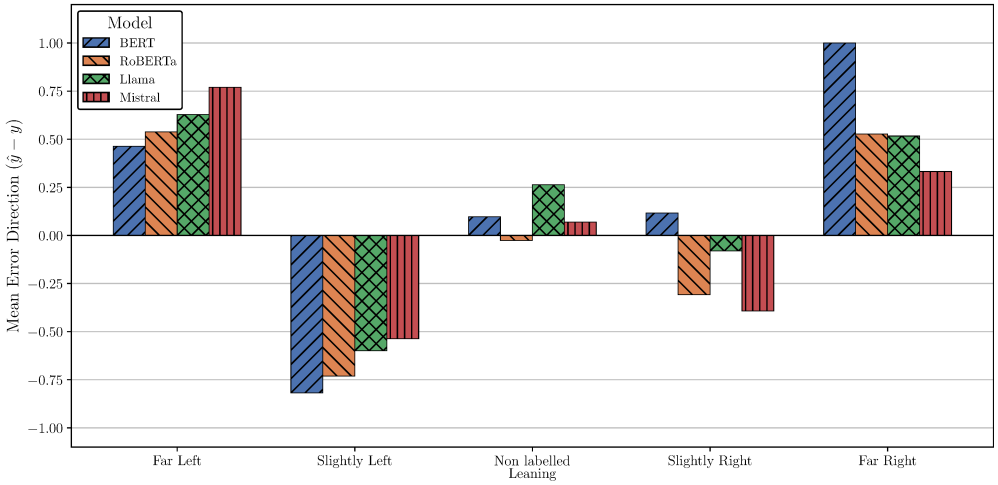


Figure 6: Mean Error Direction by political leaning. Positive values indicate overestimation of reliability, negative values underestimation.

5 Political Leaning: Distribution and Error Rates

Figure 5 gives the reliability composition of each leaning and the per-leaning error rates; Figure 6 gives the Mean Error Direction by leaning discussed in the main paper.

6 Robustness of the Bias and Performance Analyses

Leaning and reliability are publisher attributes shared by all articles of an outlet. The analyses below repeat the tests at the publisher level and control for reliability.

Contingency tables. Table 1 gives the tables behind the article-level χ^2 tests of the main paper. With two outcomes, Cramér's $V = \sqrt{\chi^2/N}$.

Independence at the publisher level. With one observation per publisher (whether its aggregated prediction is correct), the association holds for every model, with no expected count below five (Table 2).

	BERT		RoBERTa		Llama 3		Mistral	
Leaning	correct	wrong	correct	wrong	correct	wrong	correct	wrong
Far Left	299	716	288	727	285	730	370	645
Slightly Left	1018	1389	1167	1240	1269	1138	1044	1363
Non labelled	4292	2130	4376	2046	4340	2082	4463	1959
Slightly Right	5062	2428	5123	2367	4976	2514	5166	2324
Far Right	678	1464	798	1344	809	1333	869	1273
Total	11349	8127	11752	7724	11679	7797	11912	7564

Table 1: Political leaning against classification outcome, article level (19,476 articles per model).

Model	n	χ^2	p**-value**	**Cramér's V**
BERT	439	44.84	4.3×10^{-9}	0.320
RoBERTa	439	18.83	8.5×10^{-4}	0.207
Llama 3	439	23.88	8.4×10^{-5}	0.233
Mistral	439	24.06	7.8×10^{-5}	0.234

Table 2: Political leaning vs. classification correctness, one observation per publisher, 4 degrees of freedom.

Controlling for reliability. Table 3 reports a logistic regression of the article-level error on political leaning, with the true reliability level as a control and standard errors clustered on the publisher. Far-left and far-right outlets keep 4.0 to 6.5 times the odds of error of non-labelled ones. The wide intervals reflect sparse leaning-by-reliability cells (Table 5).

Uncertainty on publisher-level scores. Table 4 gives 95% intervals from a bootstrap that resamples publishers. The intervals of the four models overlap, so their differences are within sampling variation.

Leaning and reliability in the corpus. Table 5 counts publishers per combination. Of the 207 slightly right publishers, 142 sit in l_1; no slightly left publisher is in l_1 and no far-right publisher in l_5. Table 6 gives the error rate inside each combination: the two largest blocks are classified well (5% and 12% error), every other combination falls between 32% and 99%.

The per-leaning disparity is therefore a property of the corpus and the model together. A system trained on this ground truth inherits the political structure of the dataset, whatever the mechanism.

7 Experimental Details

Folds. All articles of a publisher (up to 200) stay in the same fold. Across folds, the share of each reliability class has a standard deviation of 1.37 to 3.61 percentage points.

Optimisation and batching. Tokenizers: `BertTokenizerFast`, `RobertaTokenizerFast`, and the causal `LlamaTokenizer` and `MistralTokenizer` with manual padding; Flash Attention 2 for Mistral. Batch sizes (training/evaluation): 32/16 for BERT, 16/8 for RoBERTa, 8/8 for Llama with gradient accumulation, 6/48 for Mistral with five accumulation steps. Only Llama 3 uses early stopping, on a stratified 10% hold-out of the training articles (patience two epochs), restoring the state with the lowest validation loss; the other models run for a fixed

Model	Leaning	Odds ratio	95% CI	p
BERT	Far Left	4.81**	[1.74, 13.34]	0.003
	Slightly Left	2.11*	[1.10, 4.05]	0.025
	Slightly Right	2.11*	[1.05, 4.23]	0.035
	Far Right	6.52***	[2.18, 19.47]	0.001
RoBERTa	Far Left	5.55**	[1.69, 18.26]	0.005
	Slightly Left	1.49	[0.63, 3.51]	0.360
	Slightly Right	2.46*	[1.18, 5.11]	0.016
	Far Right	6.18**	[1.97, 19.37]	0.002
Llama 3	Far Left	6.29**	[1.86, 21.28]	0.003
	Slightly Left	1.41	[0.60, 3.33]	0.436
	Slightly Right	2.74*	[1.23, 6.10]	0.014
	Far Right	6.29**	[1.93, 20.56]	0.002
Mistral	Far Left	4.00*	[1.13, 14.13]	0.031
	Slightly Left	2.12	[0.89, 5.08]	0.090
	Slightly Right	2.56*	[1.18, 5.53]	0.017
	Far Right	5.57**	[1.81, 17.13]	0.003

Table 3: Odds ratios of misclassification by political leaning, relative to the non-labelled group, controlling for the true reliability level; standard errors clustered on the publisher. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Model	Accuracy [95% CI]	Tolerant acc. α=1.0 [95% CI]
BERT	0.665 [0.617, 0.708]	0.870 [0.838, 0.900]
RoBERTa	0.690 [0.647, 0.731]	0.875 [0.843, 0.904]
Llama 3	0.665 [0.620, 0.708]	0.879 [0.847, 0.907]
Mistral	0.683 [0.642, 0.727]	0.856 [0.822, 0.888]

Table 4: Publisher-level metrics with cluster-bootstrap confidence intervals (2,000 resamples of the 439 publishers).

number of epochs. No test fold informs training or model selection. A fixed seed (42) gives identical splits across models.

8 Convergence Analysis

In a preliminary 20-epoch run of Llama 3 (Figure 7a), the validation loss is lowest at epoch 4 (0.4416) and the generalisation gap widens afterwards, so the other models are trained for five epochs. For Mistral (Figure 7b), the training loss averaged over folds falls from 2.023 to 0.355 and 0.231 in the first three epochs, so it is trained for three.

9 Reproducibility

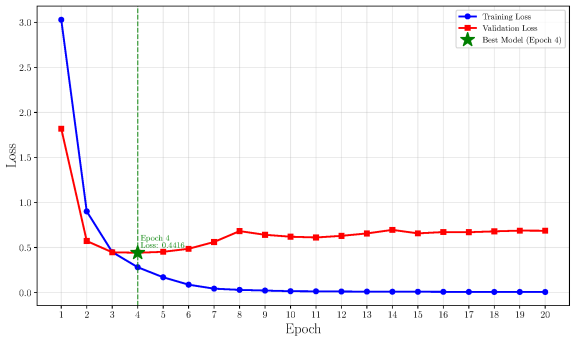
A single notebook in the accompanying code regenerates every figure and table of this document and of the main paper from the stored predictions, and checks that all models are scored against the same ground truth.

	Far Left	Sl. Left	Non lab.	Sl. Right	Far Right
l_1	1	—	2	142	7
l_2	6	7	5	22	29
l_3	1	4	64	19	7
l_4	9	29	24	17	9
l_5	3	10	15	7	—
Total	20	50	110	207	52

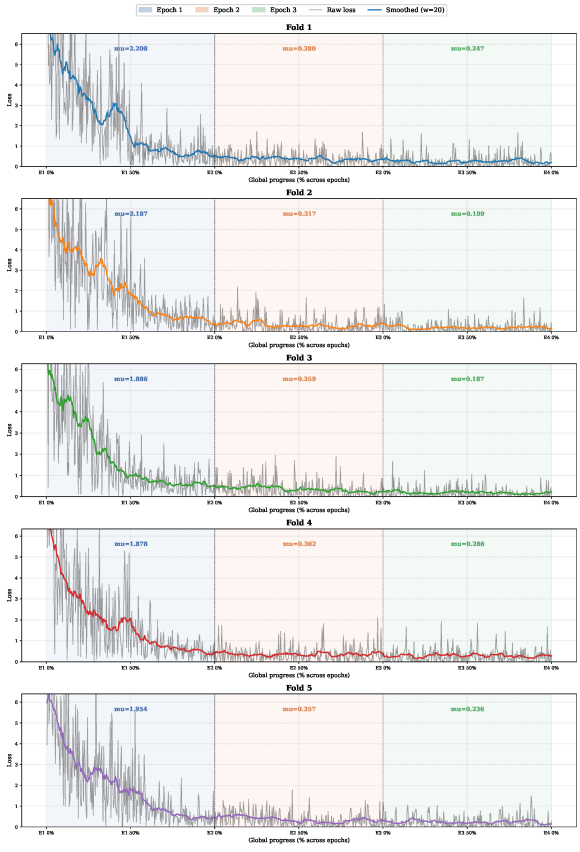
Table 5: Publishers by political leaning and true reliability level. A dash marks an empty combination.

	Far Left	Sl. Left	Non lab.	Sl. Right	Far Right
l_1	99	—	81	**5**	82
l_2	76	48	91	40	44
l_3	79	93	**12**	79	90
l_4	48	39	32	68	70
l_5	94	73	69	87	—

Table 6: Article-level error rate (%) by political leaning within each true reliability level, pooled over the four models. Bold: the two combinations that hold most of the corpus.



(a) Training and validation loss over 20 epochs for Llama 3; the green marker is the best checkpoint (epoch 4).



(b) Training loss over three epochs for Mistral, across the five folds.